run on different JVMs, either on the same host or on different machines, the messages must be converted from objects into bytes and vice versa. Let’s look at how to serialize them.

You can add various libraries to help with this or create your own serializer. The most common option, recommended for its good performance and easy setup, is an Akka module that uses Jackson serialization. It comes with two serializers to choose from. To use it, add its dependency in build.sbt:

```
val AkkaVersion = "2.6.20"

libraryDependencies += "com.typesafe.akka" %% "akka-serialization-jackson"
➥% AkkaVersion
```

Then add one of these two serializers to your application configuration:

```
akka {
  actor {
    serializers {
      jackson-json = "akka.serialization.jackson.JacksonJsonSerializer"  ❶
      jackson-cbor = "akka.serialization.jackson.JacksonCborSerializer"  ❷
    }
  }
}
```

❶ Serializes to JSON

❷ Serializes to a compact binary form

These two are included in reference.conf, which is included in the `akka-serialization-jackson` library to provide the default configuration for this library. `JacksonCborSerializer` provides a binary format that is more compact and has slightly better throughput than `JacksonJsonSerializer`, which is text-based. To select either format, set the binding as follows: